GutGlow: An Artefact-Aware Multimodal Wearable Framework for Continuous Crohn's Disease Activity Monitoring

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Abstract

Existing monitoring methods for Crohn's Disease (CD) are episodic, invasive, and reliant on clinical visits; therefore, the methods are unable to track the rapid inflammatory transitions that occur between visits. The proposed solution is called GutGlow, this framework utilizes an awareness of the user's physical environment and blends multiple sources of data, including amount of physical activity, skin temperature, frequency of bowel patterns, and sleep efficiency to provide continuous estimates of activity of Crohn's Disease. A synthetic longitudinal dataset was created from 50 simulated subjects monitored over 90-days (a total of 4,500 subject-days). The dataset included probabilistic estimations of flares and the dynamic nature of inflammatory markers (including C-reactive protein [CRP], fecal calprotectin [FCP], and Crohn's Disease Activity Index [CDAI]) as well as confounding physiological artefacts (e.g., febrile illness, behaviour-related sleep deprivation). Multi-signal consistency scoring and temporal smoothing via a rolling window were employed in the Random Forest classifier processing prior to generating the final output. The classifier identified area under receiver operating characteristic curve (AUC-ROC) = 0.984, sensitivity = 0.905, specificity = 0.973, and negative predictive value (NPV) = 0.992 from the held-out testing data set (n= 1,350; 7.8% flare prevalence). Spearman's correlations indicated that the four domains were clinically relevant in terms of signal-probable biomarker associations (ρ = 0.119-0.189, all $p<0.001$). In terms of feature importance, sleep efficiency had the greatest predictive power (51.8%), followed by consistency score (22.9%). The study findings indicate that multimodal wearable technology designed to account for artefacts may be a feasible monitoring mechanism for CD and support future clinical validation.

Keywords: Crohn's disease; inflammatory bowel disease; wearable sensor; continuous monitoring; Random Forest; multimodal fusion; artefact detection; sleep efficiency

Introduction

Crohn's disease (CD) is a persistent, long-term inflammatory illness affecting the intestines and belongs to the category of inflammatory bowel diseases (IBD) along with ulcerative colitis. (Loftus, 2004) The total number of patients worldwide suffering from IBD exceeds 6.8 million and continues to grow, especially in the newly industrialized countries of the world. (Ng et al., 2017) Crohn's is described as transmural inflammation of the intestinal wall, having multiple sites of infection along the intestines, and having a fluctuating or unstable course or cycle; resulting in clinical features such as bowel discomfort & cramping, diarrhea, ongoing fatigue & low level of physical fitness from poor nutrition. (Torres et al., 2017)

A significant issue for the management of CD is the ability to predict how quickly the disease may progress; delayed recognition of a flare is associated with more hospitalization and less quality of life. (Cosnes et al. 2011) The main ways of monitoring disease activity is by using the following tools: colonoscopy, laboratory blood tests (elevated levels of C-reactive protein, elevated levels of sedimentation rate), fecal calprotectin, and clinical disease activity scores (Crohn's Disease Activity Index & Harvey-Bradshaw Index). (Best et al., 1976; Harvey & Bradshaw, 1980) These tools are used intermittently, are often painful; and for a high percentage of patients, the clinical scores displayed do not correlate to the presence of active disease seen on endoscopy. (Peyrin-Biroulet et al., 2015)

Consumer-level tech is seeing advancements in wearable technology that allow us to monitor a person's body without having to be invasive to them continuously. A majority of studies suggest that systemic inflammation resulting from Crohn's Disease (and related diseases) can be measured in the data generated from wearables; Fatigue caused by inflammation can result in less ambulation, (Hügle et al., 2021), pyrogenic cytokines increase temperatures at the extremities (Moheet et al., 2017), nocturnal GI symptoms lead to sleep fragmentation (Ranjbaran et al., 2007), and an increased rate of bowel movements can result in changes in behaviour (Sherwood et al., 2012). However, the challenge around the accuracy and specificity of wearables and their ability to assist in the detection of disease exacerbation, as many things can cause similar variances in data; fever, physical and emotional stress and work-life balance can each cause similar variances in data separate from the act of experiencing a flare, (Lonini et al., 2018).

Based on the issues outlined above we developed GutGlow a wearable system designed to assist in the understanding of flare detection by providing artefact awareness through the use of multiple physiological domain data, smoothing of temporal data and consistency scoring between domains. Both authors (Anshul R. and Ajay Panchal) worked collaboratively to develop the Random Forest classification pipeline and the artefact aware consistency model. The primary aim of developing the GutGlow was to determine if it would be feasible to estimate the presence of a flare in a Crohn's disease patient continually while exposed to noise created through data simulation.